The King's Diagonal

Counting Lattice Paths Across the Chessboard

A Pedagogical Introduction to the Delannoy Numbers

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Abstract

We pose a simple chessboard question: in how many ways can a king cross the board diagonally, using only forward moves? From this we develop the full combinatorial machinery of the Delannoy triangle. The exposition is aimed at sixth-form and early undergraduate students. We construct the triangle from a three-parent recurrence, prove a closed binomial formula, verify it against the chessboard count of 48 639, and close with the silver-mean growth rate $3 + 2\sqrt{2}$. No knowledge beyond binomial coefficients is assumed.

1 The Question

Place a king in the corner a1 of an empty chessboard. He is to walk to the diagonally opposite corner h8. To keep things finite and well-defined, we restrict his legal moves to those that make progress toward the goal:

- one square East, one square North, one square North-East (the diagonal).

He may never retreat. *How many distinct walks are available to him?*

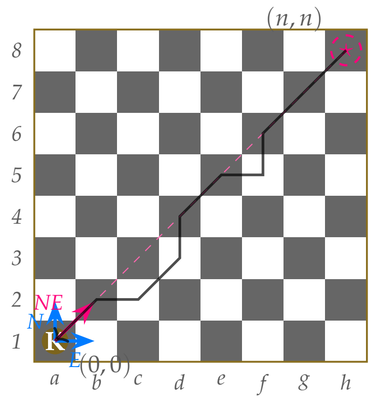

Figure 1: The king walks from a1 to h8 using only East, North, and North-East moves. Three legal first moves are shown in colour; one example completed path is drawn faintly across the board. The dashed line marks the Euclidean geodesic.

The answer though you might suspect small at 48 639, is large enough to surprise but small enough to verify by a hand-built table. Our task is to understand *why* it is that number, and to see the structure that makes the answer compressible into a few lines of arithmetic.

2 The Two-Step Cousin: Pascal's Triangle

Before introducing the diagonal, consider the simpler problem where the king moves only East or North — effectively a rook walking forward.

A rook-walk from $(0,0)$ to (n,k) consists of n North-steps and k East-steps in some order. The number of such walks is the number of ways to choose which k of the $n+k$ moves are East:

$$R(n,k) = \binom{n+k}{k}. \tag{1}$$

For the chessboard corner-to-corner traversal $(7,7)$ this gives $\binom{14}{7} = 3432$.

The values $R(n,k)$ form Pascal's triangle in rectangular coordinates, and satisfy the recurrence

$$R(n,k) = R(n-1,k) + R(n,k-1), \tag{2}$$

which says: the last step of any walk to (n,k) was either a North-step (arriving from $(n-1,k)$) or an East-step (arriving from $(n,k-1)$). Sum the two predecessor counts; you have the current count.

3 The Three-Parent Rule

Now restore the diagonal step. A walk to (n,k) may terminate in any of *three* final moves:

- East from $(n,k-1)$;
- North from $(n-1,k)$;
- North-East from $(n-1,k-1)$.

Each predecessor contributes its own count of walks. The total is their sum. This defines the **Delannoy numbers**:

$$\boxed{D(n,k) = D(n,k-1) + D(n-1,k) + D(n-1,k-1)} \tag{3}$$

with the boundary condition $D(0,0) = 1$ and $D(n,k) = 0$ whenever n or k is negative.

Along the axes, only one walk exists — straight East, or straight North — so $D(n,0) = D(0,k) = 1$. The interior is then determined entirely by (3).

4 Building the Table

Filling cells row by row from the origin gives the following grid:

$n \backslash k$	0	1	2	3	4	5	6	7
0	1	1	1	1	1	1	1	1
1	1	3	5	7	9	11	13	15
2	1	5	13	25	41	61	85	113
3	1	7	25	63	129	231	377	575
4	1	9	41	129	321	681	1289	2241
5	1	11	61	231	681	1683	3653	7183
6	1	13	85	377	1289	3653	8989	19825
7	1	15	113	575	2241	7183	19825	**48639**

The entry $D(7,7) = 48\,639$ in the bottom-right corner is our chessboard answer. We can verify a few cells by hand to confirm the rule.

- $D(1,1) = D(1,0) + D(0,1) + D(0,0) = 1 + 1 + 1 = 3$. Walks to (1,1): East-then-North, North-then-East, or one diagonal.
- $D(2,2) = D(2,1) + D(1,2) + D(1,1) = 5 + 5 + 3 = 13$.
- $D(7,7) = D(7,6) + D(6,7) + D(6,6) = 19825 + 19825 + 8989 = 48\,639$.

5 A Visual Reading

Each cell is fed by exactly three neighbours, arranged as in Figure 2.

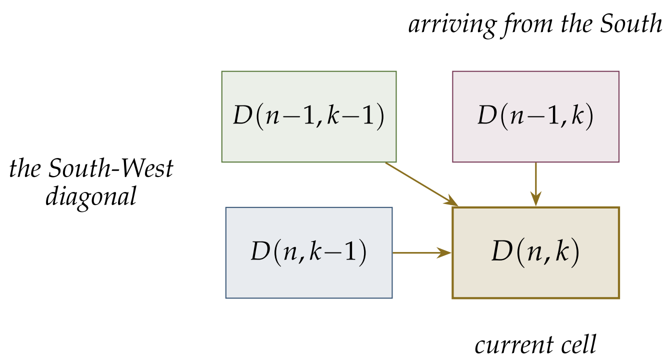

Figure 2: The three predecessors of any interior cell. Each arrow represents one of the king's legal final moves.

The rule is local. To determine $D(n,k)$ one needs only the three immediately adjacent precomputed cells. No global information is required, no backtracking. This is the same property that makes Pascal's triangle computable by elementary-school arithmetic — with the addition of one extra parent.

6 A Closed Formula

The recurrence is enough to compute any single value, but it would be more satisfying to have a closed expression.

The key observation to achieve this is that any walk to (n, k) can be classified by the number of diagonal steps it contains.

Suppose a walk uses exactly j diagonal moves. Each diagonal contributes one unit to both coordinates, so the remaining moves must contribute $k - j$ East-units and $n - j$ North-units. The total number of moves is therefore $j + (n - j) + (k - j) = n + k - j$.

To count walks with exactly j diagonals: choose which j of the $n + k - j$ slots are diagonals $\binom{n+k-j}{j}$, then choose which of the remaining $n + k - 2j$ slots are East-steps $\binom{n+k-2j}{k-j}$. Multiplying and summing over all valid j:

Proposition 1. *For $n, k \geq 0$,*

$$D(n,k) = \sum_{j=0}^{\min(n,k)} \binom{n+k-j}{j} \binom{n+k-2j}{k-j}. \tag{4}$$

Remark. An equivalent and often more convenient form is

$$D(n,k) = \sum_{j=0}^{\min(n,k)} \binom{n}{j} \binom{k}{j} 2^j,$$

obtained by reorganising the choice as "pick the j diagonals from the n North-units and the k East-units, each diagonal merging one of each." The factor 2^j tracks the freedom to interleave.

Verification at the chessboard

Apply the formula with $n = k = 7$:

$$D(7,7) = \sum_{j=0}^{7} \binom{7}{j}^2 2^j.$$

The seven terms are tabulated below.

j	$\binom{7}{j}^2$	2^j	contribution
0	1	1	1
1	49	2	98
2	441	4	1764
3	1225	8	9800
4	1225	16	19600
5	441	32	14112
6	49	64	3136
7	1	128	128
		total	**48639**

The closed formula and the recurrence agree.

7 Why So Much Larger Than Pascal?

The rook-only walk count is $\binom{14}{7} = 3432$. The king's count is 48 639. The ratio is

$$\frac{48\,639}{3432} \approx 14.2.$$

The diagonal step does not merely add a third option at the origin; it compounds. Every interior cell has three predecessors instead of two, and the counts feed forward multiplicatively. To estimate the long-run growth, consider the central diagonal $D(n,n)$:

n	$D(n,n)$	ratio to previous
0	1	—
1	3	3.00
2	13	4.33
3	63	4.85
4	321	5.10
5	1683	5.24
6	8989	5.34
7	48 639	5.41
8	265 729	5.46

The ratios are settling toward a limit. That limit is

$$3 + 2\sqrt{2} \approx 5.8284, \tag{5}$$

which is the square of the *silver mean* $1 + \sqrt{2}$. This number plays the role here that 4 plays for Pascal's triangle: it is the dominant growth rate of the recurrence.

Where the silver mean comes from

To find the asymptotic growth rate of $D(n,n)$, treat the recurrence as a linear operator on sequences and look for its dominant eigenvalue. A standard saddle-point analysis of the generating function

$$\sum_{n,k\geq 0} D(n,k)\, x^n y^k = \frac{1}{1-x-y-xy}$$

shows that along the diagonal $n = k$, the dominant singularity sits at $x = y = \sqrt{2} - 1$, giving growth rate $(\sqrt{2}-1)^{-2} = 3 + 2\sqrt{2}$.

Less formally: the silver mean satisfies $\alpha^2 = 2\alpha + 1$, which is the same algebraic relation that governs the cells along the central diagonal of the Delannoy table. The recurrence is, in disguise, a Pell-type recursion — the cousin of Fibonacci that arises whenever a structure is built on "two-plus-a-diagonal" rather than "two-plus-nothing."

8 Closing Thought

The chessboard problem has the cleanest possible answer: *three parents, one rule, build the table.* Every cell is just the local sum of its three predecessors, and the global count emerges by recursion. The same logic gives $D(2,2) = 13$ by hand and $D(7,7) = 48\,639$ by table, with no algebraic trickery.

What makes the Delannoy numbers worth attention beyond this exercise is the arithmetic that grows on top of the geometry: the central values $1, 3, 13, 63, 321, 1683, \ldots$ also count king-walks across a chessboard, appear as evaluations of Legendre polynomials at the value 3, satisfy non-trivial congruences modulo primes, and share their generating function with the Schrödinger numbers from one-dimensional walk theory.

But all of that begins with the same observation we made in section 3: three parents, one rule. The rest is just following the consequences.

Exercise. A bishop on an empty board moves only diagonally. If we allow the bishop to step North-East, South-East, or stay diagonal in some sense — find a sensible variant of the question — what triangle does the bishop's walk generate? *Hint: think about parity colouring.*

Exercise. Compute $D(10, 10)$ in two ways: by extending the table, and by Proposition 1. Confirm both give 8 097 453.

Exercise. Show directly from the closed formula that $D(n, k) = D(k, n)$. What property of king-walks does this symmetry reflect?